Practice Problems #17

1. True or False. Justify your answer.

   Let $V = \mathbf{R^2}$, which we showed is a vector space with the usual operations

   $$u + v = (u_1 + v_1, u_2 + v_2), \quad cu = (cu_1, cu_2).$$

   Then,

   $$H = \{(x, y) : xy \leq 0\},$$

   is a subspace of V.

2. Show $c0 = 0$ for every scalar c following the proof below. Justify each step.
   Proof: $c0 = c(0+0) =$ ____________________ $+$ ____________________.
   Then,
   $0 = -c0 + c0 = (-c0 + c0) +$ ____________________ $= 0 + c0 =$ ____________________.

3. For $u \in V$, where V is a vector space, suppose

   $$\alpha u = 0$$

   for some nonzero scalar α. Show that $u = 0$. Justify each and every step in your proof.

Hints or Answers are on the next page.

#1: False. Why?

#2: Mimic what we did in class.

#3: Note that $u = 1u$ (which axiom?). Use the result from #2.